Response to the Editor's Comments

General comment

Comment: "Based on the reviewer's comments received, significant revisions need to be made to further consider this manuscript for publication. The reviewers highlighted the need to improvise methodological descriptions, clarify the advancement compared to all previous relevant literature, and align the discussion and conclusions closely with the evidence presented in the study."

Response: We thank the Editor for this observation. In response to the comments received, the manuscript was comprehensively restructured, with substantial improvements to the methodology, the contextualization of the study within the relevant literature, and the alignment between the results, discussion, and conclusions. The interpretations were also revised to avoid statements that exceeded the evidence obtained, and the main limitations of the study were explicitly incorporated.

Because the restructuring affected virtually the entire manuscript, conventional track changes did not provide a clear representation of the revisions. Therefore, we have submitted the clean version requested by the journal, together with an additional version in which the revised or newly added text is highlighted in red to facilitate evaluation.

1. Sequence trimming, alignment parameters and morphometric variability

Comment: "Further include 1) additional details on sequence trimming, alignment parameters, and morphometric variability that would improve reproducibility"

Response: The molecular methodology was expanded to provide a more detailed description of sequence processing and alignment. In addition, morphometric variability is presented using the mean, standard deviation, and range for the evaluated developmental stages, thereby improving the reproducibility and interpretation of the data.

Location: page 06, lines 165–208.

2. Transmission experiments or pathogen detection analyses

Comment: "Clarify missing transmission experiments or pathogen detection analyses"

Response: It was clarified that the study did not include direct transmission experiments or pathogen detection analyses. Accordingly, the interpretations were restricted to the damage and associations directly evaluated in this study.

3. Statistical analysis

Comment:"clarify the missing statistical analysis"

Response: The statistical analysis section was expanded to specify the analyses performed, model validation procedures, significance level, and the software and packages used.

Location: page 7, line 222.

4. Disease vector status and frosty pod rot

Comment: "conclusions regarding disease-vector status and facilitation of frosty pod rot should be interpreted more cautiously because direct transmission was not assessed"

Response: We agree. The conclusions that could attribute to *M. dissimulatum* a demonstrated role as a disease vector or direct facilitator of disease were removed.

Location: page 16, lines 458–472.

5. Number of sequenced specimens

Comment: "include discussion for using small number of sequenced specimens for Molecular confirmation."

Response: This limitation was explicitly incorporated into the Discussion, clarifying that the three sequences obtained constitute complementary evidence supporting the morphological identification rather than a comprehensive assessment of the species' molecular variability.

Location: page 15, lines 435–438.

Data availability

Comment: "According to our data availability policies (...) submission of data used in the analysis either as supplementary information or to a community-endorsed, public repository is mandatory. Hence, we ask that you submit the data generated in this study to a suitable repository, and add a 'Data Availability Statement' (...)"

Response: The data generated and analyzed in this study are publicly available, and the corresponding "Data Availability" section was added to the manuscript.

Location: page 17, lines 494–497.

Inhouse Editor Comments

1. Figure 1

Comment: "Please can you clarify how the map in figure FIGURE NUMBER 1 was generated? Please cite the software used to create the map in the figure legend (including version number and URL link) if you or one of your co-authors created it."

Response: The legend of Figure 1 was revised to specify the cartographic data sources used, and information on the software employed to generate the map, including its version and URL, was added.

Location: Figure 1, page 4, lines 102–112.

2. Title

Comment: "To aid our readers, and to maximize the accessibility of your manuscript, the title should have a clear, precise scientific meaning and should not contain a punctuation (full stops, hyphen, semi-colons, colons), and also it should not exceed for more than 20 words. Where possible, the title should be read as one concise sentence. Please could you re-write the title ensuring that it is informative and appropriate."

Response: The title was reformulated to make it more concise and informative and to comply with the 20-word limit and the requirement to avoid punctuation marks.

Revised title: "Morphological, molecular, and biological characterization of *Monalonion dissimulatum* Distant associated with cacao in Amazonas, Peru"

Code availability

Comment:"Please note that if your manuscript uses any custom or bespoke computational tool or code (...) the underlying code must be deposited in a recognised DOI-assigning repository (...)"

Response: The statistical analyses were performed using R scripts. To ensure reproducibility, these scripts were deposited in a public repository with a DOI, and a "Code Availability" section was added to the manuscript.

Response to Reviewer 1

We thank the reviewer for the careful evaluation of the manuscript and for the constructive comments, which substantially contributed to improving its methodological clarity, scope, and interpretation.

Major Comment 1
Comment: "Scientific novelty should be more clearly established."
Response: The Introduction was restructured to more clearly define the knowledge gap and distinguish the contribution of the present study from previous work. In addition, the rationale for evaluating the relationship between incidence and elevation was clarified, and the study objectives were reformulated.
Location: page 3, lines 38–90.

Major Comment 2
Comment: "The molecular component should be presented in proportion to its scope."
Response: We agree. The terminology used to describe the molecular component was moderated, and it was clarified that the three COI sequences provide complementary evidence supporting the morphological identification rather than a comprehensive molecular characterization of the population. This limitation was explicitly incorporated into the Discussion.
Location: page 15, lines 428–447.

Major Comment 3
Comment: "The ecological interpretation of altitude should be more cautious."
Response: The interpretation of the relationship between elevation and incidence was revised to present it strictly as an association rather than a causal relationship. It is also acknowledged that elevation explained only 38.5% of the observed variation and that other unmeasured environmental variables may contribute to the observed pattern.
Location: page 15, lines 397–408.

Major Comment 4
Comment: "Several methodological details should be expanded to improve reproducibility."
Response: The Materials and Methods section was substantially expanded to include the plot selection criteria, management and crop characteristics, evaluation period, morphometric procedures, PCR and sequencing conditions, parameters used for sample size calculation, and details of the statistical analyses and software employed.
During the revision, a calibration problem was also identified in the equipment initially used for the morphometric measurements. Therefore, a new collection of specimens was conducted, and all measurements were repeated using a properly calibrated stereomicroscope. The data presented in the revised version correspond exclusively to these new measurements.
Location: pages 4–7, lines 91–232.

Major Comment 5
Comment: "Several datasets remain largely descriptive despite their analytical potential."
Response: The quantitative presentation was strengthened by incorporating the mean, standard deviation, and range for the morphometric variables and developmental stage durations. Fruit damage was retained as a qualitative assessment, consistent with the study design and scope.
Location: pages 11 and 13, lines 294 - 366.

Major Comment 6
Comment: "The taxonomic section would benefit from a stronger analytical focus."

Response: The taxonomic section was reorganized to place greater emphasis on characters useful for the identification of *M. dissimulatum*, complemented by comparisons with previous descriptions and with other species of the genus.
Location: pages 9–13, lines 260–366.

Major Comment 7
Comment:"The relationship between *M. dissimulatum* and frosty pod rot should be interpreted more cautiously."
Response: We agree. Statements that could be interpreted as evidence of *Moniliophthora roreri* transmission or vector competence by *M. dissimulatum* were removed.

Major Comment 8
Comment:"Several interpretations extend beyond the evidence presented."
Response: The Discussion and Conclusions were revised to restrict the interpretations to the evidence obtained. Potential ecological and management implications are now presented more cautiously and clearly distinguished from the findings directly supported by the study.
Location: pages 14–15, lines 388–472.

Major Comment 9
Comment:"The manuscript would benefit from an explicit discussion of its limitations."
Response: The main limitations of the study were incorporated into the Discussion, including the limited number of sequenced specimens, the observational nature of the relationship between incidence and elevation, the absence of measurements of other environmental variables, the qualitative assessment of fruit damage, and the absence of pathogen transmission experiments.

Moderate Comment 1
Comment: "Comparisons with previous studies could be developed further."
Response: The Discussion was expanded to compare the morphological, molecular, and biological results with previous studies and to consider possible explanations for the observed differences. The discussion of the use and limitations of the COI marker in Miridae and *Monalonion* was also strengthened.
Location: page 15, lines 388–456.

Minor Editorial Comments
Comment: "Several editorial issues should also be corrected during revision. The legend of Figure 2 refers to 'disease incidence' instead of insect incidence. Figure 4 would benefit from identifying which structures are taxonomically diagnostic. Table 1 should include measures of variation, while Table 2 would be more informative if mean developmental times were reported in addition to ranges. The duplicated paragraph describing necrotic lesions in the Discussion should be removed, the typographical error 'his study' (line 324) should be corrected to 'This study', citation formatting should be standardized throughout the manuscript, and the Conclusions should avoid introducing concepts that were not experimentally evaluated."
Response: The editorial issues identified were corrected. The tables now include measures of variation and mean values, the figure legends and figures were revised, redundancies and typographical errors were removed, and the Conclusions were reformulated to remain limited to the aspects directly evaluated in the study.

Response to Reviewer 2

We thank the reviewer for the comments and suggestions, which contributed to improving the organization, taxonomic foundation, and methodological clarity of the manuscript.

Comment 1: "The manuscript is not in a logical sequence, many things that should be in the introduction are in the results or discussion, some of the methods in the results, etc."
Response: The manuscript was comprehensively restructured to improve its logical flow. The content was reorganized across the Introduction, Materials and Methods, Results, and Discussion sections to avoid overlap between sections.

Comment 2: "The authors don't provide enough information in the introduction about the species subject of study. Furthermore they are missing key papers like those of Carvalho."
Response: The Introduction was expanded to provide greater taxonomic and biological context for *M. dissimulatum*. In addition, the fundamental taxonomic works by Carvalho and recent relevant literature on the genus *Monalonion* were incorporated.
Location: page 3, lines 38–90.

Comment 3: "The authors use indistincly Monalonion as a genus or as a species throughout the manuscript, this needs to be clear."
Response: The entire manuscript was revised to clearly distinguish references to the genus *Monalonion* from those specifically referring to *M. dissimulatum*, ensuring consistent use of taxonomic nomenclature throughout.

Comment 4: "Although the authors claims they ¨confirm¨ identification by using molecular data, they only are matching the data with previously well identified specimens. Also, there are few problems with molecular identification and though discussion in the last years which the authors omit and use relatively old references from a time when the COI identifaction was on boom."
Response: We agree. The interpretation of the molecular component was moderated, presenting the COI sequences as complementary evidence supporting the morphological identification rather than as independent taxonomic confirmation. In addition, the Discussion was updated to acknowledge the limitations of the COI marker and the small number of sequenced specimens.
Location: pages 13 and 15, lines 367–376 and 428–447.

Comment 5: "There is no formal description of every nymphal instar, which in a work like this needs to be done, and ideally a key to these instars."
Response: A differentiated description of each of the five nymphal instars was incorporated, considering body size, wing-pad development, and the main external characteristics that allow recognition of their ontogenetic progression.
Location: pages 9–11, lines 265–294.

Comment 6: "There is no clarification on how the observed specimens in the field were identified, if there are cryptic species the data might be biased."

Response: The specimen identification methodology was clarified, indicating that taxonomic assignment was based on diagnostic morphological characters and that a subset of specimens was additionally evaluated using COI sequences as complementary evidence. It is also acknowledged that the limited molecular sampling does not allow genetic variation or cryptic diversity within the evaluated populations to be comprehensively ruled out.
Location: pages 9–15, lines 260–380 and 428–433.

Response to Reviewer 3

We thank the reviewer for the comments and suggestions, which helped improve the clarity of the figure and the overall presentation of the manuscript.

Comment 1: "Figure 1. The map turned out very well, but it does not include the names of all the districts such as La Peca, mentioned in the text above. I was unable to identify the dashed line indicating the provincial boundary. I suggest ensuring the colors match those shown in the map legend; the same should be done for the other elements."
Response: Figure 1 was revised to include the names of the districts mentioned in the text, improve the identification of the provincial boundary, and ensure consistency between the colors and symbols shown on the map and those in the legend.
Location: page 4, lines 92–112.

Comment 2: "Lines 344–354 and 355–365. Duplicate paragraph."
Response: The duplicated paragraph was removed, and the corresponding section was revised to avoid redundancy.